Shiftless 3-adic Collatz (Collatz-3) Model:

Ternary Interference and Carry Avalanches

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1. Introduction

Building upon prior work, the 3-adic Collatz (Collatz-3) map in this study applies the following operations:

- For non-multiples of 3 (*i.e.*, $n \equiv 1, 2 \ (mod\ 3)$: $(4n \pm 1)/3$
- For multiples of 3: $n/3$

In base-3, division by 3 is essentially a right shift, an operation that explicitly erases the least significant trit. This loss of information makes the internal structure of the trajectory unobservable and obscures the geometry of carry propagation. To eliminate this information loss, we propose the Shiftless Collatz-3 model, an information-preserving variant of the Collatz-3 map. Its update rule is defined as:

$$n_{k+1} = 4n_k + c_k \cdot 3^{v_k}$$

Here, v_k is the 3-adic valuation ($v_3(n_k)$), equal to the index of the lowest non-zero trit (NZT). Each n_k admits a unique factorization:

$$n_k = 3^{v_k} \cdot u_k$$

where u_k (the non-3 part) matches exactly the transitions of the standard 3-adic Collatz sequence.

Depending on the NZT, the following rule ensures that the lowest trit is deterministically zeroed out while preserving the full ternary history:

- $\mathrm{NZT} = 1 \ \rightarrow \ c_k = -1$
- $\mathrm{NZT} = 2 \ \rightarrow \ c_k = +1$

2. Equivalence to the Standard 3-adic Collatz Model

The Shiftless Collatz-3 model is mathematically equivalent to the standard Collatz-3 map (which includes right shifts). Under the unique decomposition

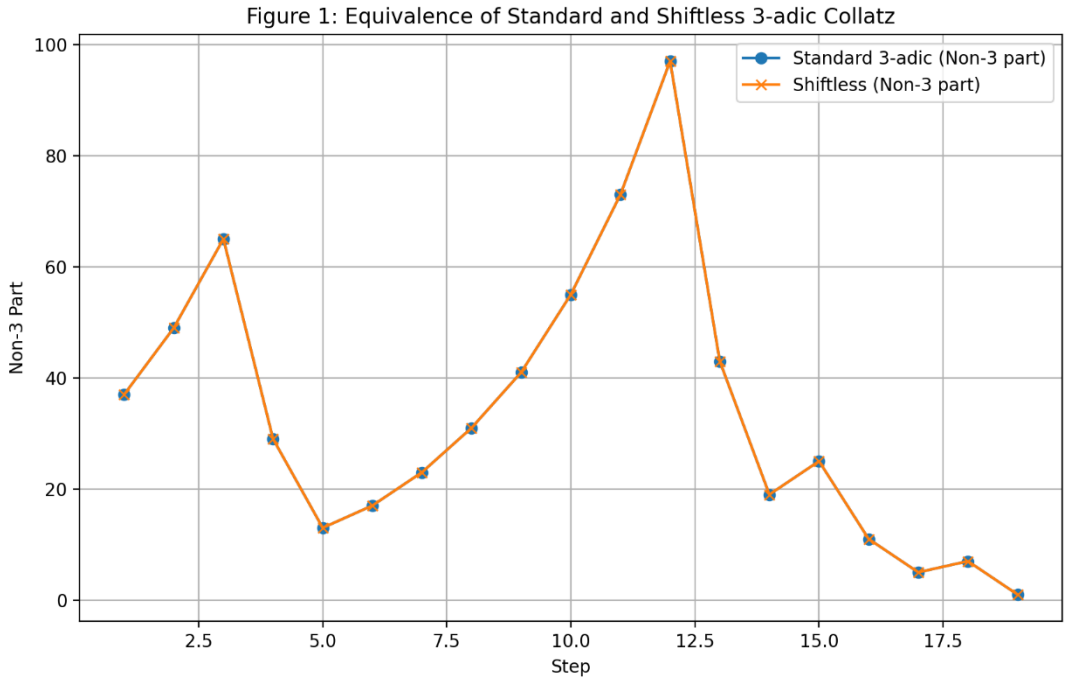
$$n_k = 3^{v_k} \cdot u_k$$

the non-3 part (u_k) follows exactly the same transitions as in the standard model.

In particular, the following properties coincide:

- The number of steps required to reach a jackpot (3^m)
- The sequence of transitions of the non-3 part

Thus, the Shiftless Collatz-3 model is an information-preserving equivalent of the standard Collatz-3 map. **Figure 1** illustrates the case ($n_0 = 28$).

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Verification for 3-adic Seed: 28
Step  | Standard (Non-3)   | Shiftless                | Non-3(Shiftless)   | Match?
-------------------------------------------------------------------------------------
1     | 37                 | 111                      | 37                 | True
2     | 49                 | 441                      | 49                 | True
3     | 65                 | 1755                     | 65                 | True
4     | 29                 | 7047                     | 29                 | True
5     | 13                 | 28431                    | 13                 | True
6     | 17                 | 111537                   | 17                 | True
7     | 23                 | 452709                   | 23                 | True
8     | 31                 | 1830519                  | 31                 | True
9     | 41                 | 7263027                  | 41                 | True
10    | 55                 | 29229255                 | 55                 | True
11    | 73                 | 116385579                | 73                 | True
12    | 97                 | 463947993                | 97                 | True
13    | 43                 | 1851009003               | 43                 | True
14    | 19                 | 7360989291               | 19                 | True
15    | 25                 | 29056536675              | 25                 | True
16    | 11                 | 115063885233             | 11                 | True
17    | 5                  | 470715894135             | 5                  | True
18    | 7                  | 1977006755367            | 7                  | True
19    | 1                  | 7625597484987            | 1                  | True
-------------------------------------------------------------------------------------
Verification Successful. Total steps to Jackpot: 19
Final Jackpot Value: 3^27
```

Figure 1. Comparison of the ternary non-3 parts of the standard and the shiftless Collatz-3 sequence (n = 28).

3. Shiftless Collatz-3 Trajectory and Carry Avalanches

3.1 Compact Trajectory

A defining feature of the Shiftless model is the intense carry avalanche triggered when $\mathrm{NZT} = 2$ ($i.e.,\ c_k = +1$).

Example:

$$4 \times 2 + 1 = 9 = 100_3$$

This reveals:

- The lowest two trits are simultaneously zeroed out
- Carries propagate upward in a chain reaction
- An asymmetric carry structure inherent to the ternary digit system emerges

These avalanches expose a ternary gasket structure far more intricate than its binary counterpart. Regions with consecutive $\mathrm{NZT} = 2$ exhibit higher-order fractal carry interference (**Figure 2**).

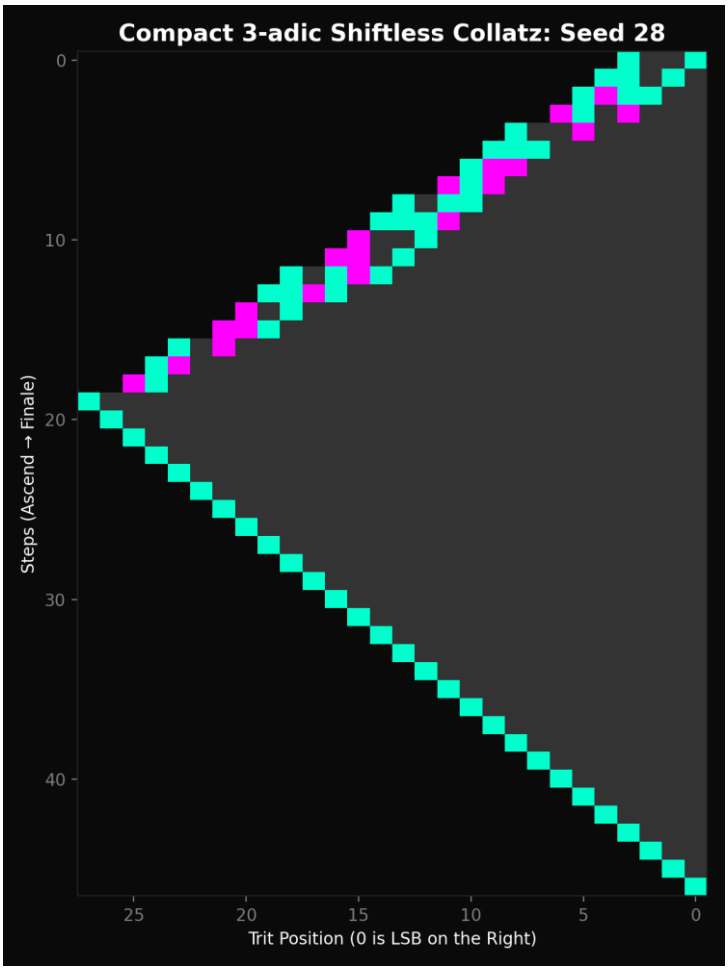

Figure 2. Compact orbit visualization of the shiftless Collatz-3 trajectory (*n* = 28).

3.2 Full Trajectory

Each step of the Shiftless Collatz-3 trajectory decomposes into two phases:

Reach: Corresponds to a left shift + self-superposition in ternary notation, isomorphic to Rule 90 gasket generation.

$$n \mapsto 4n$$

Fill: Injection of the NZT term introduces local distortion and triggers a carry avalanche.

$$4n \mapsto 4n + c_k \cdot 3^{v_k}$$

Figure 3 visualizes the full trajectory.

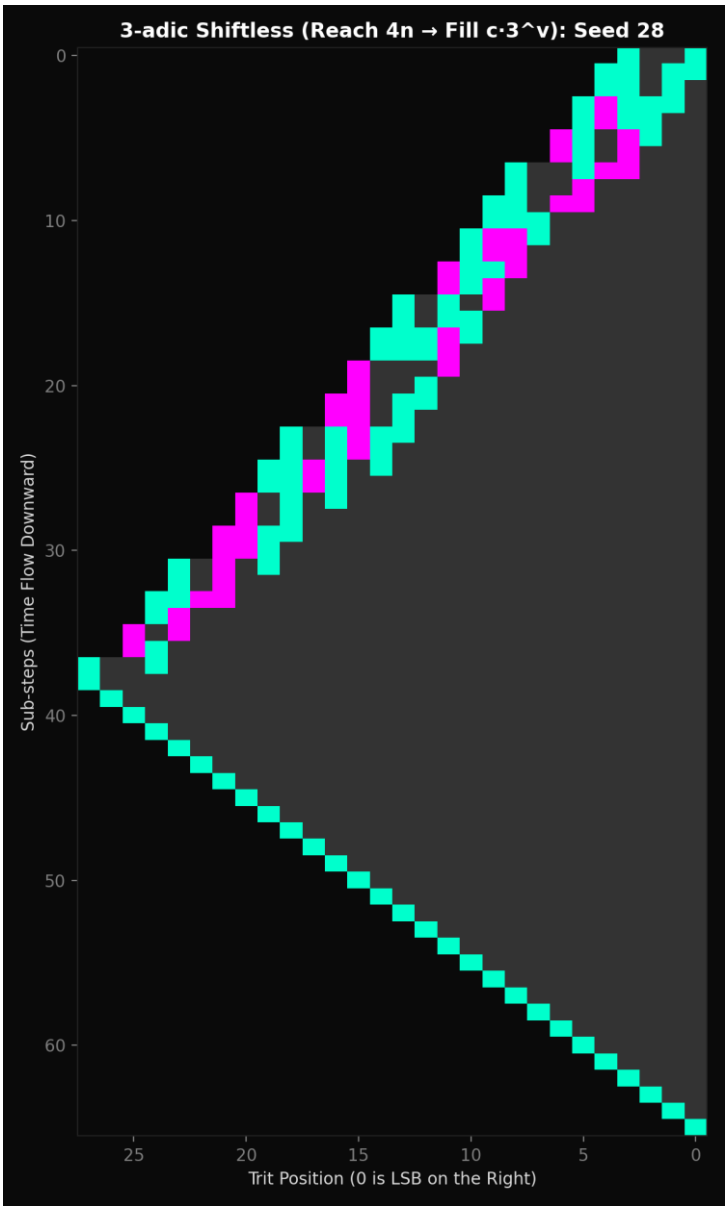

Figure 3. Full orbit visualization of the shiftless Collatz-3 trajectory (*n* = 28).

3.3 Comparison of Three Dynamics

Using the initial value $n_0 = 28$, we compare the following three dynamics:

- $4n$ — pure geometric expansion
- $4n + 1$ — simple noise injection
- $4n + c_k \cdot 3^{v_k}$ — Shiftless Collatz-3

Results (**Figure 4**):

- $4n$ exhibits a clear ternary gasket
- $4n + 1$ is slightly distorted but macroscopically resembles the pure expansion
- $4n + c_k \cdot 3^{v_k}$ undergoes structural breakdown due to the heavy NZT injection

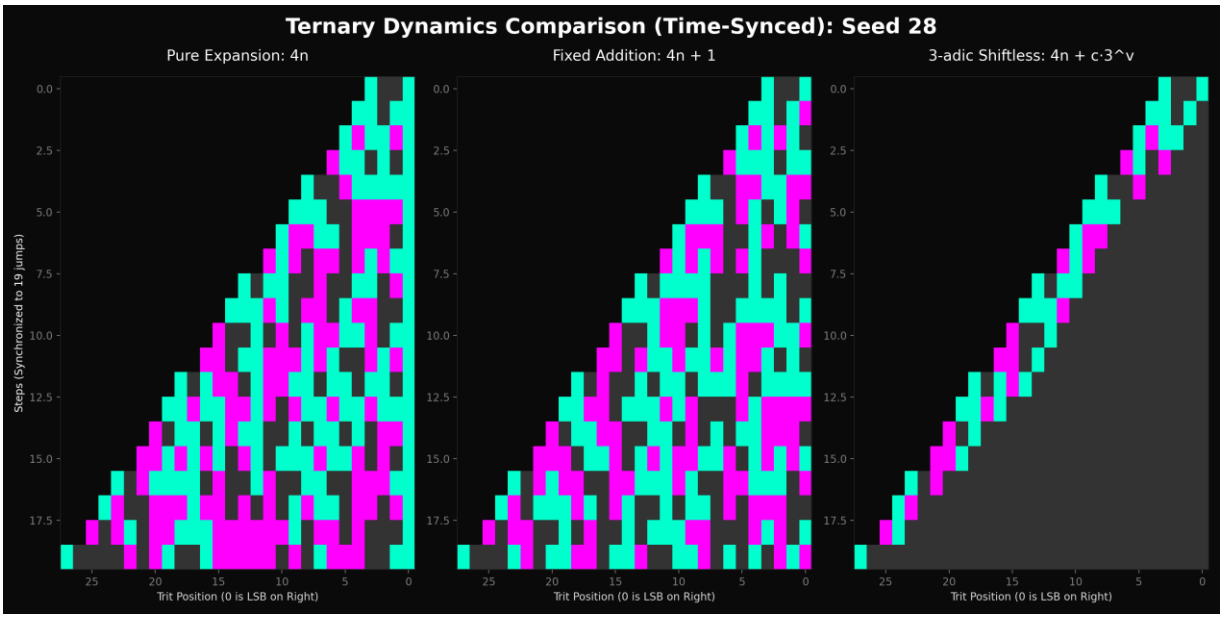

Figure 4. Synchronized comparison of the three dynamics: 4*n*, 4*n* + 1, and 4*n* + *NZT* (*n* = 28).

4. Linear Decomposition and Interference Geometry

4.1 Linear Decomposition

The Shiftless Collatz-3 trajectory admits a perfect linear decomposition:

$$n_k = 4^k n_0 + \sum_{i=0}^{k-1} 4^{k-1-i} C_i$$

$$C_i = c_i \cdot 3^{v_i}$$

Term A: Pure Expansion

$$A_k = 4^k n_0$$

- No carries
- No interference
- Represents the "order" of the 3-adic structure

Term B: Complement Interference

$$B_k = \sum_{i=0}^{k-1} 4^{k-1-i} C_i$$

- Encodes NZT sign information
- Source of carry avalanches
- Produces ternary moiré (interference fringes)

4.2 Geometry of Interference

When Term A and Term B have non-zero trits at the same position, ternary addition triggers:

$$\text{carry} \rightarrow \text{annihilation of higher trits}$$

This cascading mechanism is the geometric origin of the carry avalanche (**Figure 5**).

5. Conclusion

The Shiftless Collatz-3 model provides a powerful framework that:

- Is equivalent to the standard Collatz-3 map
- Avoids information loss from right shifts
- Visualizes the geometry of carry propagation
- Enables linear decomposition of trajectories

It exhibits fractal dynamics absent in 2-adic systems, driven by:

- Asymmetric carry avalanches inherent to ternary digits
- Strong interference between Term A and Term B

Worldview contrast:

- 2-adic Shiftless: Noise floats on a sea of order (A dominates)
- 3-adic Shiftless: Order sinks in a sea of noise (B dominates)

This difference arises because the base p determines both the intensity of carry avalanches and the granularity of noise (the weight of NZT injections).

Future directions include:

- Measure-theoretic analysis
- A unified theory with the Shiftless 2-adic model
- Generalization to p-adic Collatz structures

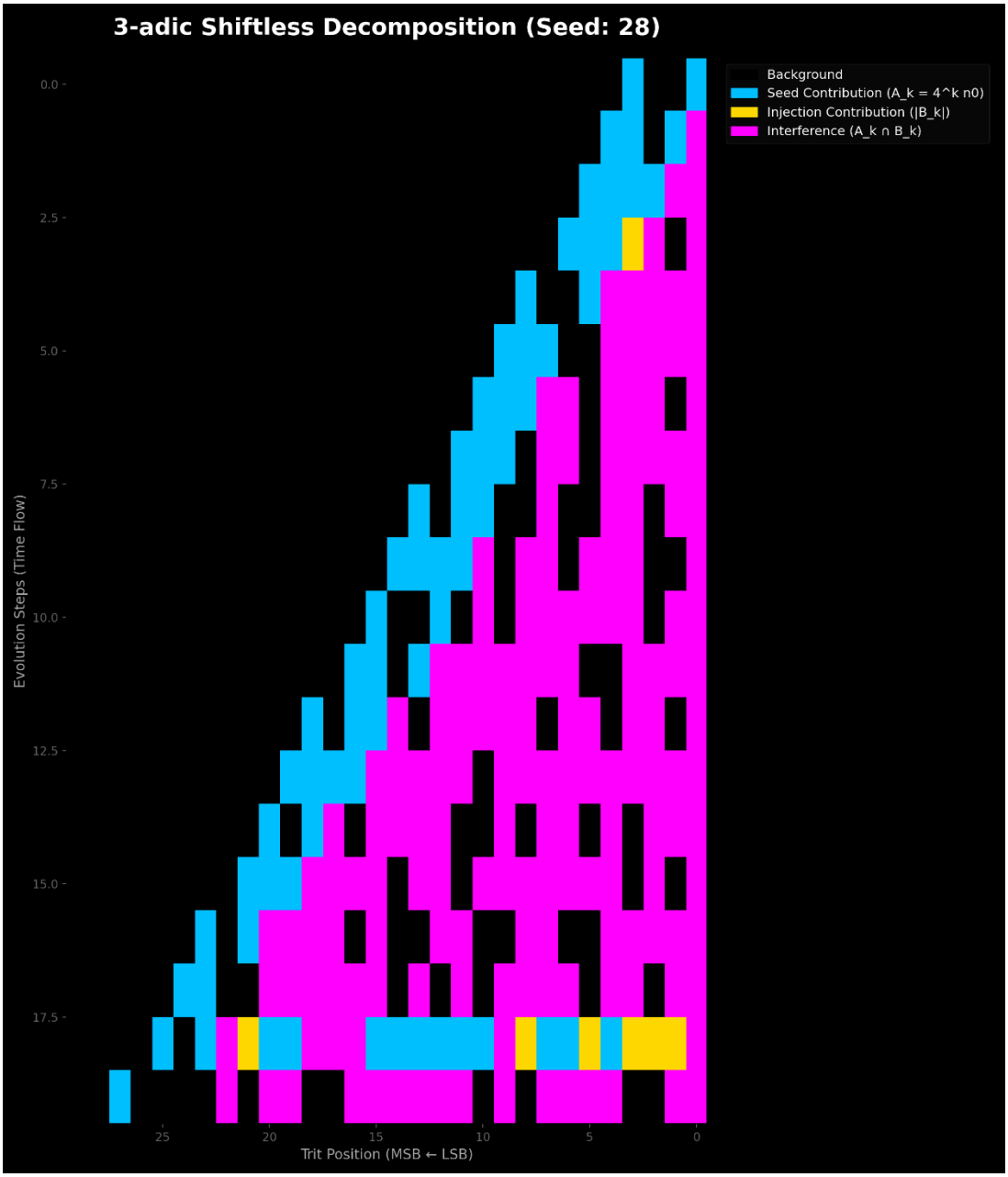

Figure 5. Trit-wise visualization of the linear decomposition of the sequence n_k into A_k and B_k (n = 28).

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